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Education

Rey Juan Carlos University **Sep. 2023 – Present**
Double Degree in Mathematics and Software Engineering (4th year) *Madrid, Spain*

- Expected graduation: June 2028. Selected grades: Design & Analysis of Algorithms (9.8/10), Probability (9.7/10), Data Structures (9.1/10).
- **Core Coursework**: Algorithms & Complexity, Data Structures, Operating Systems, Probability, Linear Algebra, Discrete Mathematics, Formal Languages, Differential Equations.

Academica International Studies **Sep. 2020 – June 2022**
US High School Diploma (Dual Diploma Program), graduated with Honors *Miami, USA (Remote)*

IES Villa de Valdemoro **Sep. 2021 – May 2023**
Spanish Baccalaureate, Bachillerato de Excelencia (high academic performance program) *Madrid, Spain*

Projects

TikZfy: Image-to-TikZ Neural Engine | *PyTorch, FastAPI, Astro, Docker, GCP* | GitHub **Jun. 2026 – Sep. 2026**

- Image-to-TikZ model: 57.8M-parameter CoordConv encoder and causal transformer decoder, with prefix-trie constrained decoding.
- Holdout evaluation at 256×256: 100.0% of outputs compile with TeX Live, mean SSIM 0.919 (cross-entropy baseline: 57.1% and 0.342). The 128×128 CoordConv model already compiled 100%; the final model improves SSIM (0.732 to 0.919) and family accuracy (99.3%).
- Trained on 240,000 synthetic diagram pairs (41 GB of memory-mapped shards) with a three-stage curriculum and composite loss, on one NVIDIA L4 (Google Cloud).
- Ports-and-adapters service: PyTorch core, TeX Live compilation through a subprocess adapter, FastAPI endpoint, Astro frontend, Docker; 279 tests and CI (Ruff, typecheck, pytest).

Conway's 99-Graph: formal and SAT results | *Lean 4, Python, CaDiCaL, drat-trim* | Zenodo **Sep. 2026 – Oct. 2026**

- Lean 4 proof with no `sorry`, axioms `[propext, Quot.sound]`: no automorphism of order 7 for graphs with parameters $srg(99, 14, 1, 2)$.
- Lean 4 reduction of the involution case (choice-free): an involution fixing exactly one vertex must have no twin matching edge. The remaining case is stated explicitly (`noInvolution_iff_q0`); it is not proved.
- DRAT proof of the order-7 case (24.9 GiB), checked with `drat-trim` (`s VERIFIED`), archived on Zenodo (DOI 10.5281/zenodo.22995101).
- Case split of the remaining involution case into 1,010 subcases: 16 cubes refuted with drat-trim-verified proofs; 16 open after 20–35 h of CaDiCaL, Kissat and ParaFROST runs with no verdict.
- Bounded CaDiCaL search on order-3 and involution branches (12.5 days on GCP, over 2.4×10^9 conflicts); these branches are not decided.
- The existence of the graph remains open.

MCP-Manim-Slides | *Python, MCP SDK, FFmpeg, Playwright* | GitHub **Aug. 2026 – Sep. 2026**

- Open-source MCP server (MIT license) that lets coding agents write, render and compile Manim-Slides presentations: 12 tools, 5 prompt templates and 3 resources.
- Per-scene render cache keyed by content hash; `sync_deck` re-renders only the scenes that changed.
- Export to Reveal.js, PDF, PPTX and MP4 (FFmpeg); Playwright screenshots for visual checks. Tests with pytest and linting with Ruff in CI.

Monte Carlo Option Pricing Engine | *C++, pybind11, CMake, Python* | GitHub **Jun. 2026 – Aug. 2026**

- Multithreaded C++ Monte Carlo pricer for European options: per-thread `std::mt19937_64` generators and per-thread partial sums, exposed to Python through pybind11; built with `-O3 -ffast-math -march=native`.
- Convergence tests against the closed-form Black–Scholes price, and a memory-growth test on the C++ engine. Notebooks for finite-difference Greeks and the implied-volatility smile.

Algorithmic Trading Backtester | *Python, NumPy, pandas, pytest* | GitHub **Jun. 2026 – Aug. 2026**

- Vectorized backtesting engine (NumPy, pandas) with fixed and proportional transaction-cost models behind a common interface.
- Signals are shifted by one bar before execution to avoid look-ahead bias. Momentum and SMA-crossover strategies included.
- Unit tests check the Sharpe ratio, maximum drawdown and annualized volatility against their definitions, and the friction models against hand-computed costs.

Technical Skills

Languages: Python, C++, Java, SQL, Lean 4, Bash, TypeScript, R, MATLAB
ML & Data: PyTorch (bf16 AMP, curriculum training, constrained decoding), NumPy, pandas, SciPy, scikit-learn
Systems & Tooling: FastAPI, pybind11, CMake, Docker, Git, GitHub Actions, pytest, ruff, uv, Linux, GCP
Formal Methods: Lean 4, CaDiCaL, drat-trim, DIMACS/Tseitin encodings
AI tooling: MCP server design; agent-assisted development (Claude Code, Codex)

Certifications, Activities & Languages

Certifications: Bloomberg Market Concepts (BMC); Santander Scholarship – AI & Productivity (2024).
Activities: URJC Cybersecurity League, cryptography track (Sep. 2024).
Languages: Spanish (native), English (fluent, C1; US Dual Diploma).